

## HDOC DAY-12 ACTIVITY

### QUESTION 1 – Menu-Driven Digit Operations

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Find the sum and product of the first and last digits. E.g. Number: 12345, Sum = 6, Product = 5.
- Count how many digits are lesser than, equal to, and greater than a given digit k. E.g. Number: 12345, k = 3, Count\_lesser = 2, Count\_equal = 1, Count\_greater = 2.

#### 1. Problem Understanding

- Input: A positive integer and the user's menu choice.
- For the second operation, input the digit k (0–9).
- Output: The result of the operation selected through switch-case.

#### Operation Summary

| Choice | Operation                                | Main Logic                                             |
|--------|------------------------------------------|--------------------------------------------------------|
| 1      | Sum and product of first and last digits | Find first and last digits; calculate sum and product. |
| 2      | Count digits lesser/equal/greater than k | Extract each digit and compare it with k.              |
| Other  | Invalid choice                           | Display an appropriate message.                        |

#### 2. Standard Test Cases

| Test Case | Choice | Input               | Expected Output                          | Actual Output                            | Result |
|-----------|--------|---------------------|------------------------------------------|------------------------------------------|--------|
| 1         | 1      | n = 12345           | Sum = 6;<br>Product = 5                  | Sum = 6;<br>Product = 5                  | Pass   |
| 2         | 1      | n = 9087            | Sum = 16;<br>Product = 0                 | Sum = 16;<br>Product = 0                 | Pass   |
| 3         | 2      | n = 12345,<br>k = 3 | Lesser = 2;<br>Equal = 1;<br>Greater = 2 | Lesser = 2;<br>Equal = 1;<br>Greater = 2 | Pass   |
| 4         | 2      | n = 55555,<br>k = 5 | Lesser = 0;<br>Equal = 5;<br>Greater = 0 | Lesser = 0;<br>Equal = 5;<br>Greater = 0 | Pass   |

#### 3. Algorithm

##### A. Common Menu Logic

1. Start the program.
2. Declare required variables.
3. Read the positive integer.

4. Display the menu and read the user's choice.
5. Use switch-case to perform the selected operation.
6. Display the result.
7. Stop the program.

#### **B. Case 1 – Sum and Product**

8. Find the last digit using  $n \% 10$ .
9. Repeatedly divide a copy of  $n$  by 10 until one digit remains; this is the first digit.
10. Calculate  $\text{sum} = \text{first digit} + \text{last digit}$ .
11. Calculate  $\text{product} = \text{first digit} \times \text{last digit}$ .
12. Display the sum and product.

#### **C. Case 2 – Digit Comparison**

13. Initialize the three counters to zero and read  $k$ .
14. Extract each digit using  $n \% 10$ .
15. Compare the digit with  $k$  and increment the appropriate counter.
16. Remove the last digit using  $n / 10$  and repeat.
17. Display the three counts.

## **5. C Program**

```
#include <stdio.h>

int main(void)
{
    int n, choice, k, digit, temp;
    int first, last, sum, product;
    int count_lesser, count_equal, count_greater;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    printf("\nMenu\n");
    printf("1. Find sum and product of first and last digits\n");
    printf("2. Count digits less than, equal to, and greater than k\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1:
            last = n % 10;
            temp = n;
```

```
while (temp >= 10)
temp = temp / 10;
first = temp;
sum = first + last;
product = first * last;
printf("Sum = %d\n", sum);
printf("Product = %d\n", product);
break;
case 2:
printf("Enter digit k (0-9): ");
scanf("%d", &k);
count_lesser = 0;
count_equal = 0;
count_greater = 0;
while (n > 0)
{
digit = n % 10;
if (digit < k)
count_lesser++;
else if (digit == k)
count_equal++;
else
count_greater++;
n = n / 10;
}
printf("Count_lesser = %d\n", count_lesser);
printf("Count_equal = %d\n", count_equal);
printf("Count_greater = %d\n", count_greater);
break;
default:
printf("Invalid choice.\n");
}
```

```
return 0;
```

```
}
```

## 6. Compilation and Execution Evidence

```
kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
$ gedit question.c
** (gedit:4490): WARNING **: 11:48:03.830: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:4490): WARNING **: 11:48:03.830: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
(kali@kali)-[~/Desktop]
$ gcc question.c -o question
(kali@kali)-[~/Desktop]
$ ./question
Enter a positive integer: 12345
Menu
1. Find sum and product of first and last digits
2. Count digits less than, equal to, and greater than k
Enter your choice: 1
Sum = 6
Product = 5
(kali@kali)-[~/Desktop]
$ ./question
Enter a positive integer: 9087
Menu
1. Find sum and product of first and last digits
2. Count digits less than, equal to, and greater than k
Enter your choice: 1
Sum = 16
Product = 63
(kali@kali)-[~/Desktop]
$ ./question
Enter a positive integer: 12345
Menu
1. Find sum and product of first and last digits
2. Count digits less than, equal to, and greater than k
Enter your choice: 2
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
Product = 5
(kali@kali)-[~/Desktop]
$ ./question
Enter a positive integer: 9087
Menu
1. Find sum and product of first and last digits
2. Count digits less than, equal to, and greater than k
Enter your choice: 1
Sum = 16
Product = 63
(kali@kali)-[~/Desktop]
$ ./question
Enter a positive integer: 12345
Menu
1. Find sum and product of first and last digits
2. Count digits less than, equal to, and greater than k
Enter your choice: 2
Enter digit k (0-9): 3
Count_lesser = 2
Count_equal = 1
Count_greater = 2
(kali@kali)-[~/Desktop]
$ ./question
Enter a positive integer: 55555
Menu
1. Find sum and product of first and last digits
2. Count digits less than, equal to, and greater than k
Enter your choice: 2
Enter digit k (0-9): 5
Count_lesser = 0
Count_equal = 5
Count_greater = 0
(kali@kali)-[~/Desktop]
$
```

## 7. Verification of All Test Cases

| Test Case | Choice | Input               | Expected Output                          | Actual Output                            | Result |
|-----------|--------|---------------------|------------------------------------------|------------------------------------------|--------|
| 1         | 1      | n = 12345           | Sum = 6;<br>Product = 5                  | Sum = 6;<br>Product = 5                  | Pass   |
| 2         | 1      | n = 9087            | Sum = 16;<br>Product = 0                 | Sum = 16;<br>Product = 0                 | Pass   |
| 3         | 2      | n = 12345,<br>k = 3 | Lesser = 2;<br>Equal = 1;<br>Greater = 2 | Lesser = 2;<br>Equal = 1;<br>Greater = 2 | Pass   |
| 4         | 2      | n = 55555,<br>k = 5 | Lesser = 0;<br>Equal = 5;<br>Greater = 0 | Lesser = 0;<br>Equal = 5;<br>Greater = 0 | Pass   |